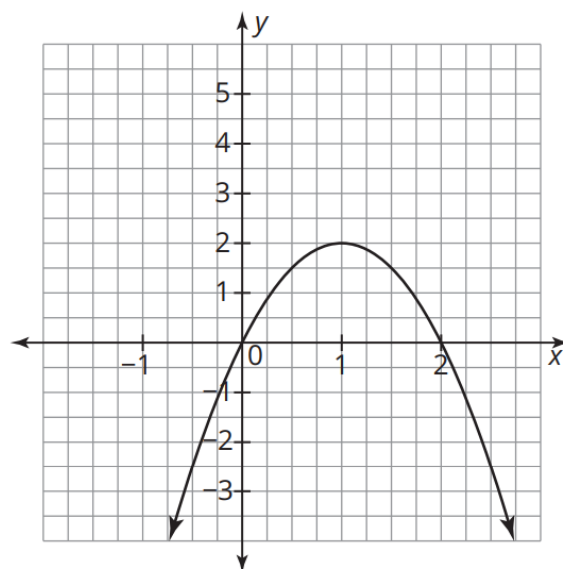


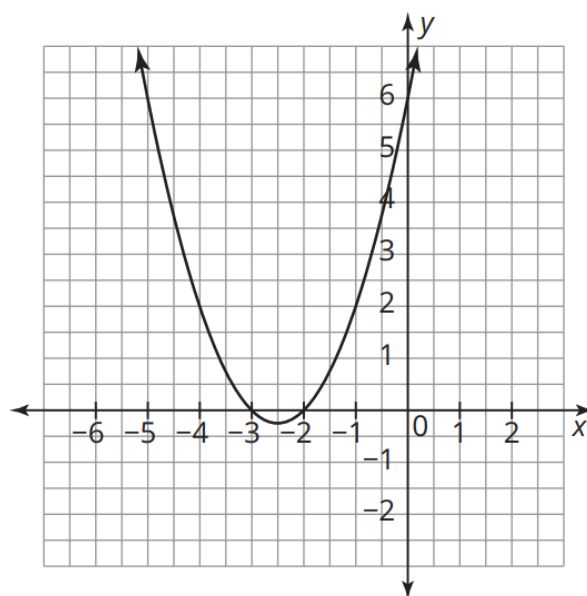
Characteristics of Quadratic Functions**Example 1:**

For each function shown, identify the domain, range, x-intercepts, y-intercept, axis of symmetry, vertex, and interval of increase and decrease.

a. The graph shown represents the function $f(x) = -2x^2 + 4x$.

Domain:**Range:****x-intercepts:****y-intercept:****Axis of symmetry:****Vertex:****Interval of increase:****Interval of decrease:**

b. The graph shown represents the function $f(x) = x^2 + 5x + 6$.

Domain:**Range:****x-intercepts:****y-intercept:****Axis of symmetry:****Vertex:****Interval of increase:****Interval of decrease:**

Example 2: Use your graphing calculator to find the zeros of the functions and then write the equation in factored form.

A) $r(x) = -2x^2 + 6x + 20$

B) $w(x) = -x^2 - 4x$

C) $h(x) = x^2 - 8x + 12$

D) $w(x) = 3x^2 - 3$